Deducing acceleration

AS Level | 9702 AS Ch.2 | Chapter 2: Accelerated motion | Lesson 2.4 | Sections 2.4

Goal: Find acceleration from the gradient of a velocity–time graph.

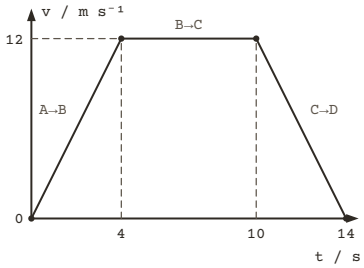
01 The big picture

On a velocity–time graph the gradient is $\Delta v / \Delta t$, which is the acceleration. A steeper line means a larger acceleration; a horizontal line means constant velocity (zero acceleration); a line sloping down means the object is slowing.

A straight line means uniform acceleration. For a curved line, draw a tangent at the moment you want and find its gradient with a large triangle.

Gradient [paraphrase]
Change in velocity ÷ change in time, read from a large triangle on the line.

Fig. 1 · A velocity-time graph



Steeper line = larger acceleration. Flat = constant velocity. Downward slope = slowing down.

02 Key equations

Acceleration = gradient of v-t

$$a = \frac{\Delta v}{\Delta t}$$

Symbols and units

v	velocity	$\mathrm{m\,s^{-1}}$	t	time	s	a	acceleration	$\mathrm{m\,s^{-2}}$

03 Sample problems with solutions

Sample 1 · Reading the graph
Use the graph in Fig. 1 to find the acceleration during each of the three stages.

Step 1. A to B:

$$a = \frac{12 - 0}{4 - 0} = 3.0\ \mathrm{m\,s^{-2}}$$

Step 2. B to C: the line is horizontal, so the velocity is constant: $a = 0$.

Step 3. C to D: the gradient is negative, so it is slowing:

$$a = \frac{0 - 12}{14 - 10} = -3.0\ \mathrm{m\,s^{-2}}$$

Answer: +3.0 m s^{-2}, then 0, then −3.0 m s^{-2}

Unit check: $\frac{\mathrm{m\,s^{-1}}}{\mathrm{s}} = \mathrm{m\,s^{-2}}$ ✓

04 Practice problems

P1 What does a horizontal line on a velocity–time graph show?
Answer: Constant velocity (zero acceleration)

P2 A straight line goes from (0 s, 4.0 m s^{-1}) to (5.0 s, 14 m s^{-1}). Find the acceleration.
Answer: 2.0 m s^{-2}

P3 A tangent to a curved v–t graph passes through (0 s, 2.0 m s^{-1}) and (4.0 s, 10 m s^{-1}). Find the acceleration there.
Answer: 2.0 m s^{-2}

P4 A v–t line falls from 20 m s^{-1} to 0 in 8.0 s. Find the acceleration.
Answer: −2.5 m s^{-2}

05 Other useful information

Common mistakes

✗ **Wrong:** Read a horizontal v–t line as "at rest".
✓ **Right:** That is an s–t graph. On a v–t graph it means constant velocity.

Exam tips

- Check the axes first: s–t gradient = velocity; v–t gradient = acceleration.

What the syllabus asks

- 9702 topic 2: acceleration from the gradient of a v–t graph. Coursebook 2.4.

Explore: Practice questions for this lesson